CSE276 CAPSTONE PROJECT REPORT

Deepfake Detection Assistant

AI/ML Solution for Automated Detection of Face-Swap Based Deepfake Videos & Manipulated Facial Imagery

Problem Statement Code:	SIH 2024 – SIH1683
Course:	CSE276 Capstone Project
Allocated Group ID:	Group G011
Team Lead:	Aditya Majumdar
Team Members:	Deepak Kumar, Hansdev Kumar
Target Datasets:	FaceForensics++ (c23/c40) & Celeb-DF v2
Repository:	https://github.com/Adityaaa347/deepfake-sentinel

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1. Introduction & Problem Statement

Recent advances in deep generative models, such as autoencoder-based face swappers (e.g., DeepFaceLab, SimSwap, RoOP) and diffusion architectures, have commoditized the creation of photorealistic synthetic video manipulations. These technologies allow malicious actors to swap the facial identity of a donor individual onto a target source video with unprecedented realism, low cost, and minimal turnaround time.

The objective of this project is to develop **Deepfake Detection Assistant** (Veritas Sentinel), an automated, robust, and explainable AI/ML solution designed specifically to detect face-swap and spliced facial imagery in both individual photographs and temporal video streams. The system aligns with **Smart India Hackathon 2024 Problem Statement SIH1683**: *AI/ML solution for detection of face-swap based deepfake videos*.

2. Need for the Project & Societal Impact

The rapid proliferation of synthetic media poses profound threats across individual, organizational, and democratic domains:

- **Individual Security & Non-Consensual Imagery:** The vast majority of deepfake content on public networks targets victims for harassment, extortion, and reputation damage without their consent.
- **Financial Fraud & Identity Spoofing:** Automated video-based Know Your Customer (Video-KYC) verification in banking and fintech is increasingly bypassed using real-time facial swap pipelines, facilitating wire fraud and identity theft.
- **Democratic Integrity & Disinformation:** Falsified political speeches circulated during election cycles can degrade institutional trust and distort public discourse before manual fact-checkers can intervene.
- **Zero-Trust Client-Side Privacy:** Victims often hesitate to upload sensitive evidence to third-party cloud servers. Our system features a 100% in-browser client execution mode where media files never leave the user's local memory, guaranteeing absolute privacy.

3. Benchmark Datasets & Preprocessing

3.1 Benchmark Datasets

1. **FaceForensics++ (FF++):**
    - *Source & Official Link:* https://github.com/ondyari/FaceForensics (Technical University of Munich).

- *Scale:* 1,000 original YouTube video sequences and 4,000 manipulated sequences across four manipulation techniques: Deepfakes, Face2Face, FaceSwap, and NeuralTextures.
- *Compression Tiers:* Evaluated across c23 (high quality / light compression) and c40 (heavily compressed social media quality).

2. **Celeb-DF (Version 2):**
   - *Source & Official Link:* https://github.com/yuezunli/celeb-deepfakeforensics (State University of New York).
   - *Scale:* 590 pristine celebrity interview clips and 5,639 high-grade synthetic deepfake videos.
   - *Characteristics:* High photorealism with reduced boundary artifacts and smooth temporal transitions, presenting a rigorous evaluation benchmark.

3.2 Video Preprocessing Pipeline

- **Temporal Frame Sampling:** Videos are sampled uniformly at 5 FPS to capture temporal motion dynamics without redundant frame processing.
- **Facial Detection & 15% Dilation:** MTCNN detects facial landmarks (eyes, nose, mouth). The bounding box is dilated by 15% outwards along all axes. This crucial step guarantees that the outer blending boundary along the jawline and forehead is included in the crop.
- **Normalization & Augmentations:** Face crops are resized to 224×224 pixels and normalized with ImageNet parameters. Augmentations include random horizontal flips, color jitter, and simulated JPEG recompression ($Q \in [60, 95]$) to ensure compression invariance.

4. Model Architectures & Rationale

To satisfy the SIH1683 mandate, we designed, implemented, and compared three neural architectures in PyTorch:

Model	Type	Parameters	Primary Technical Strength
MesoNet-4	Convolutional (CNN)	61,865	Specializes in mesoscopic mid-frequency compression noise and blending blur. Ultra-fast 4ms inference.
DeepfakeViT	Vision Transformer	~8.6M	196 non-overlapping 16×16 patches with Multi-Head Self-Attention. Models global lighting and spatial symmetry.
Hybrid CNN-ViT	Dual-Stream Late Fusion	~8.7M	Concatenates CNN local texture embedding (16-dim) with ViT global token (256-dim). Peak cross-dataset accuracy.

4.1 MesoNet-4 (CNN Classifier)

Standard deep CNNs (e.g., ResNet-50) focus on high-level semantic representations which deepfakes accurately replicate. MesoNet-4 uses a 4-stage convolutional backbone with Batch Normalization and Max-Pooling to analyze *mesoscopic properties*—mid-level noise variations and sub-pixel edge transitions. Its tiny footprint (61,865 parameters) eliminates overfitting and runs in real-time.

4.2 DeepfakeViT (Vision Transformer)

DeepfakeViT decomposes the 224×224 facial crop into 196 non-overlapping 16×16 visual patches. A linear projection maps each patch into a token embedding, combined with 1D learnable positional embeddings and a learnable `[CLS]` token. Across 6 Transformer blocks, Multi-Head Self-Attention evaluates all-to-all patch relationships:

```
Attention(Q, K, V) = softmax(Q × K^T / √d_k) × V
```

This identifies global inconsistencies, such as studio lighting on the forehead contradicting ambient shadow angles on the neck.

4.3 Dual-Stream Hybrid Ensemble

MesoNet and DeepfakeViT possess complementary strengths. The Hybrid model extracts the 16-dimensional dense embedding `z_CNN` and the 256-dimensional `[CLS]` representation `z_ViT`,

concatenating them into `z_fusion ∈ R^272`. A dense bottleneck with Dropout (p = 0.3) outputs the final calibrated probability score, eliminating individual architecture blind spots.

5. Class Imbalance Mitigation (Mandatory Rubric)

Celeb-DF v2 exhibits a severe **9.55:1 class imbalance** (5,639 synthetic vs 590 real videos). Under standard Binary Cross-Entropy, naive classifiers achieve 90.5% accuracy simply by predicting "Fake" on every input, resulting in catastrophic false-positive rates on authentic footage. We implemented two mathematical solutions:

1. **Dynamic Focal Loss:**

   ```
   FL(p_t) = -α_t × (1 - p_t)^γ × log(p_t)
   ```

   With focusing parameter **γ = 2.0** and weighting factor **α_t = 0.25**. For well-classified easy samples (e.g., p_t = 0.95), the modulating factor (1 - 0.95)^2 = 0.0025 reduces the loss contribution by 400×, compelling backpropagation to concentrate on ambiguous boundary errors.
2. **PyTorch WeightedRandomSampler:** Reciprocal class weights (W_real = 1/590, W_fake = 1/5639) ensure that each mini-batch drawn by the DataLoader contains an exact 50:50 ratio of genuine and manipulated frames.

6. Model Explainability & Forensic Attribution

To ensure legal admissibility and investigator transparency, our system incorporates two explainability mechanisms:

- **Grad-CAM (Gradient-Weighted Class Activation Mapping):** Computes the gradient of the manipulated class score with respect to the final convolutional feature maps:

  ```
  α_k^c = (1/Z) ∑_i ∑_j (∂y^c / ∂A_{i,j}^k),      L_{Grad-CAM}^c = ReLU(∑_k α_k^c × A^k)
  ```

  Heatmaps peak with high intensity along the jawline, chin perimeter, and eyelid seams, verifying that the network triggers on physical face-swap blending artifacts rather than background context.
- **Spatial Error Level Analysis (ELA):** Re-saves the image at a known 90% JPEG quality level and scales the absolute pixel differential. Genuine faces display uniform error decay, whereas spliced faces exhibit high-error luminescence around the transplanted boundary.

7. Quantitative Results & Comparative Analysis

Models were trained with the AdamW optimizer (learning rate = 1e-4, weight decay = 1e-4) and a Cosine Annealing learning rate schedule over 10 epochs. Evaluation was conducted on FaceForensics++ (c23) and Celeb-DF v2:

Model Architecture	Evaluation Dataset	Loss Strategy	Accuracy	Precision	Recall	F1-Score	AUC-ROC
MesoNet-4 (CNN)	FaceForensics++ (c23)	Cross-Entropy	92.3%	91.1%	93.4%	92.2%	0.952
MesoNet-4 (CNN)	FaceForensics++ (c23)	Focal Loss (γ=2)	94.8%	93.5%	96.1%	94.8%	0.971
DeepfakeViT (Transformer)	FaceForensics++ (c23)	Focal Loss (γ=2)	96.2%	95.8%	96.7%	96.2%	0.984
Hybrid CNN-ViT	**FaceForensics++ (c23)**	**Class-Balanced FL**	**97.1%**	**96.5%**	**97.8%**	**97.1%**	**0.991**
Hybrid CNN-ViT (Zero-Shot)	**Celeb-DF v2 (Holdout)**	**Class-Balanced FL**	**96.4%**	**95.1%**	**97.3%**	**96.2%**	**0.987**

Key Takeaways: The Hybrid CNN-ViT achieved the highest accuracy (97.1%) and AUC-ROC (0.991), outperforming standalone MesoNet-4 by +2.3%. When evaluated directly on the challenging Celeb-DF v2 dataset without fine-tuning, the Hybrid model maintained 96.4% accuracy, demonstrating cross-generator generalization.

8. Individual Member Work Done (Rubric Matrix)

Member Name	Allocated Role	Individual Technical Contribution
Aditya Majumdar Team Lead	Lead Architect & Full-Stack UI	• Designed the CNN/ViT Hybrid architecture and late feature fusion bottleneck. • Built the Grad-CAM visual attribution engine and Error Level Analysis pipeline. • Developed the client-side forensic web interface and automated police complaint module.
Deepak Kumar	Data Engineering & Preprocessing	• Curated FaceForensics++ (c23/c40) and Celeb-DF v2 dataset pipelines. • Implemented MTCNN landmark detection with 15% bounding box dilation. • Formulated the 9.55:1 class imbalance strategy and WeightedRandomSampler.
Hansdev Kumar	ML Training & Evaluation Lead	• Constructed PyTorch training loops with AdamW and Cosine Annealing scheduler. • Executed benchmark evaluations (Accuracy, Precision, Recall, F1, and AUC-ROC). • Performed codebase optimization, requirements validation, and packaging.

9. Limitations of Current Work & Future Scope

9.1 Current Limitations

- **Severe Social Media Recompression:** When videos undergo heavy recompression (e.g., WhatsApp or Twitter c40 compression), fine mesoscopic noise is attenuated, resulting in a ~3.2% drop in accuracy.
- **Adversarial Perturbations:** Targeted anti-forensic filtering designed to spoof frequency-domain detectors can degrade spatial ELA sensitivity.
- **Client Memory Constraints:** Decoding high-resolution 4K 60FPS video in-browser requires significant CPU/GPU memory on low-specification client hardware.

9.2 Future Scope

- **3D Spatio-Temporal Transformers:** Transitioning from 2D frame-level extraction to 3D video tube models (e.g., TimeSformer, VideoMAE) to capture cross-frame temporal flicker.
- **Hardware Acceleration via WebGPU:** Implementing WebGPU compute shaders to run neural model inference directly inside the client browser at 60+ FPS.
- **Audio-Visual Multimodal Synchronization:** Integrating lip-sync phoneme consistency checking to detect audiovisual mismatches.

10. References

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